Investigation: BAHEMA EDUCAÇÃO S.A. (CVM code 701), fiscal year 2015, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
Before 2020, the company was called BAHEMA EDUCACAO S.A..
Before 2019, the company was called BAHEMA SA.
Trading codes (2023 FCA, version 2, filed on 2023-06-26): BAHI3 (Ações Ordinárias, Bolsa, since 2018-01-26).

Document used

2015 DFP, version 1, filed on 2016-03-22; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2016-03-22).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=54269

Possible manipulation (engine v1, study 1)

Financial risk (context only)

Altman Z'' (financial risk, context only)
  Value: 11.383 (index)
  Components: X1 = 0.927; X2 = 0.109; X3 = −0.073; X4 = 5.182 (indices)
  Percentile in sector and year: 84 (n = 54)
  The company's own history: median 10.636; Tukey range (1.5 × IQR): from 4.419 to 16.645; value inside the range (index; 5 fiscal years)
  - No cut-off zone: the available source does not define zones for Z''
  - Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
  Value: F = 1/9 (7 computable)
  Signals:
    - Return on assets (ROA): 0
    - Operating cash flow on assets (CFO): 0
    - Change in ROA (ΔROA): 0
    - Cash above earnings (ACCRUAL): 0
    - Change in leverage (ΔLEVER): 0
    - Change in current ratio (ΔLIQUID): 1
    - No equity issuance (EQ_OFFER): —
    - Change in gross margin (ΔMARGIN): —
    - Change in asset turnover (ΔTURN): 0
  - No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
  - No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Highlighted in the following year's comparative: 4 (threshold of R$0.4 million, 1% of total assets of the version used). Values in R$ million:

DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −4.7 in the version used (v1, filed on 2016-03-22); −0.8 in the next DFP's comparative (v1, filed on 2017-03-27); difference of 3.9. Different value in the following year's comparative (restatement or reclassification).

BPP 2.02 Passivo Não Circulante: 3.8 in the version used (v1, filed on 2016-03-22); 2.4 in the next DFP's comparative (v1, filed on 2017-03-27); difference of −1.5. Different value in the following year's comparative (restatement or reclassification).

BPP 2.03 Patrimônio Líquido: 31.4 in the version used (v1, filed on 2016-03-22); 32.8 in the next DFP's comparative (v1, filed on 2017-03-27); difference of 1.5. Different value in the following year's comparative (restatement or reclassification).

DRE 3.11 Lucro/Prejuízo do Período: −0.7 in the version used (v1, filed on 2016-03-22); 0.8 in the next DFP's comparative (v1, filed on 2017-03-27); difference of 1.5. Different value in the following year's comparative (restatement or reclassification).

Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.

Full list: 46 accounts above the R$1,000 tolerance (group a, totals and key lines: 4; group b, reclassifications: 42), in the annex at the end.

What was known on the date

- DFP 2010, v1: 2011-03-04
- DFP 2011, v1: 2012-03-16
- DFP 2012, v1: 2013-04-01
- DFP 2013, v1: 2014-03-31
- DFP 2014, v1: 2015-03-28
- FCA 2015, v1: 2015-03-28
- FCA 2015, v2: 2015-05-04
- DFP 2015, v1: 2016-03-22
- DFP 2016, v1: 2017-03-27

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 3 failed.

- Cash flow closing balance = cash and cash equivalents on the balance sheet (ÚLTIMO): difference above the tolerance of R$2,000
- Cash flow closing balance = cash and cash equivalents on the balance sheet (PENÚLTIMO): difference above the tolerance of R$2,000
- Cash flow opening balance = cash and cash equivalents on the prior balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).

no opinion excerpt was found in the text

Auditor in the 2021 FCA: BDO RCS Auditores Independentes SS.

Auditor in the 2022 FCA: BDO RCS Auditores Independentes SS.

Auditor in the 2023 FCA: BDO RCS Auditores Independentes SS.

Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2015: Audilink & Cia. Auditores -> BKS Auditores. The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2016: BKS Auditores -> Grant Thornton Auditores Independentes Ltda. The outgoing firm audited 1

consecutive fiscal years (2015 to 2015), by the start and end dates in the FCA.
- 2019: Grant Thornton Auditores Independentes Ltda -> KPMG Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2016 to 2018), by the start and end dates in the FCA.
- 2020: KPMG Auditores Independentes -> BDO RCS Auditores Independentes SS. The outgoing firm audited 1 consecutive fiscal years (2019 to 2019), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-11T00:49:34+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: BIOMA EDUCAÇÃO S.A..

Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −4.7 in the version used, −0.8 in the comparative (difference of 3.9)
(a) BPP 2.02 Passivo Não Circulante: 3.8 in the version used, 2.4 in the comparative (difference of −1.5)
(a) BPP 2.03 Patrimônio Líquido: 31.4 in the version used, 32.8 in the comparative (difference of 1.5)
(a) DRE 3.11 Lucro/Prejuízo do Período: −0.7 in the version used, 0.8 in the comparative (difference of 1.5)
(b) DFC (indirect method) 6.02.01 Adições de Investimentos: 6.2 in the version used, 0.0 in the

comparative (difference of −6.2)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: −5.2 in the version used, 0.7 in the comparative (difference of 5.9)
(b) DFC (indirect method) 6.01.01.02 Despesas/(Receitas) Que Não Afetam o Caixa: −4.6 in the version used, −0.7 in the comparative (difference of 3.8)
(b) DRE 3.06.01.02 Outras Receitas Financeiras: 3.0 in the version used, 0.0 in the comparative (difference of −3.0)
(b) DFC (indirect method) 6.02 Caixa Líquido Atividades de Investimento: 6.7 in the version used, 3.9 in the comparative (difference of −2.8)
(b) DFC (indirect method) 6.01.01.01 Lucro Líquido do Exercício: −0.7 in the version used, 1.4 in the comparative (difference of 2.1)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: 0.5 in the version used, −1.4 in the comparative (difference of −2.0)
(b) BPP 2.02.02 Outras Obrigações: 3.4 in the version used, 1.9 in the comparative (difference of −1.5)
(b) BPP 2.02.02.02 Outros: 3.4 in the version used, 1.9 in the comparative (difference of −1.5)
(b) BPP 2.02.02.02.07 Dédicit Atuarial a Pagar: 2.0 in the version used, 0.5 in the comparative (difference of −1.5)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −0.7 in the version used, 0.8 in the comparative (difference of 1.5)
(b) DRE 3.04 Despesas/Receitas Operacionais: −2.7 in the version used, −1.3 in the comparative (difference of 1.5)
(b) DRE 3.04.05 Outras Despesas Operacionais: −4.9 in the version used, −3.4 in the comparative (difference of 1.5)
(b) DRE 3.04.05.03 Provisão para Déficit Atuarial: −2.4 in the version used, −0.9 in the comparative (difference of 1.5)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: −2.7 in the version used, −1.3 in the comparative (difference of 1.5)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: −0.1 in the version used, 1.4 in the comparative (difference of 1.5)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: −0.7 in the version used, 0.8 in the comparative (difference of 1.5)
(b) BPP 2.02.02.02.06 Outras Obrigações: 0.2 in the version used, 1.4 in the comparative (difference of 1.2)
(b) DFC (indirect method) 6.01.02.03 Variação no Passivo Circulante: 1.0 in the version used, −0.1 in the comparative (difference of −1.1)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: −1.9 in the version used, −3.0 in the comparative (difference of −1.1)
(b) DFC (indirect method) 6.03.01 Dividendos e Juros s/ Capital Próprio: −1.9 in the version used, −3.0 in the comparative (difference of −1.1)
(b) BPA 1.01.01 Caixa e Equivalentes de Caixa: 0.0 in the version used, 0.7 in the comparative (difference of 0.7)
(b) BPA 1.01.03 Contas a Receber: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.01.03.02 Outras Contas a Receber: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.01.03.02.02 Outros Créditos: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPP 2.02.02.02.03 Contribuição Social s/ o Lucro: 0.6 in the version used, 0.0 in the comparative (difference of −0.6)
(b) DFC (indirect method) 6.01.02.02 Variação no Ativo Não Circulante: 1.5 in the version used, 0.9 in the comparative (difference of −0.6)
(b) BPP 2.02.02.02.05 Parcelamento Federal-REFIS(Lei.12.996/14): 0.6 in the version used, 0.0 in the comparative (difference of −0.6)
(b) DFC (indirect method) 6.02.02 Ajustes de Avaliação Patrimonial em Controladas: 0.5 in the version used, 0.0 in the comparative (difference of −0.5)

(b) BPP 2.01.05.02.04 Outras Obrigações a Pagar: 0.4 in the version used, 0.0 in the comparative (difference of −0.4)
(b) DRE 3.04.04.02 Outras Receitas: 0.2 in the version used, 0.0 in the comparative (difference of −0.2)
(b) DRE 3.04.04.04 Reversão de Provisão: 2.6 in the version used, 2.9 in the comparative (difference of 0.2)
(b) DFC (indirect method) 6.01.02.04 Variação no Passivo Não Circulante: −1.5 in the version used, −1.8 in the comparative (difference of −0.2)
(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 0.0 in the version used, 0.1 in the comparative (difference of 0.1)
(b) BPP 2.01.01.02 Obrigações Trabalhistas: 0.0 in the version used, 0.1 in the comparative (difference of 0.1)
(b) BPP 2.01.06 Provisões: 0.3 in the version used, 0.2 in the comparative (difference of −0.1)
(b) BPP 2.01.06.01 Provisões Fiscais Previdenciárias Trabalhistas e Cíveis: 0.3 in the version used, 0.2 in the comparative (difference of −0.1)
(b) BPP 2.01.06.01.03 Provisões para Benefícios a Empregados: 0.3 in the version used, 0.2 in the comparative (difference of −0.1)
(b) BPP 2.01.05.02.01 Dividendos e JCP a Pagar: 0.1 in the version used, 0.0 in the comparative (difference of −0.1)
(b) BPP 2.01.05.02.02 Dividendo Mínimo Obrigatório a Pagar: 0.0 in the version used, 0.1 in the comparative (difference of 0.1)
(b) DFC (indirect method) 6.03.04 Reversão Dividendo Prescrito e não Reclamado: 0.0 in the version used, 0.0 in the comparative (difference of −0.0)
(b) DFC (indirect method) 6.01.02.01 Variação no Ativo Circulante: −0.4 in the version used, −0.4 in the comparative (difference of −0.0)